9	**(a)**		Using $v^2 = u^2 + 2as$ with $s = 0.8, u = 6, a = 9.8$ $v^2 = 6^2 + 2 \times 9.8 \times 0.8$ $v = \sqrt{51.68} = 7.19$ m s^{-1}	**M1** **A1** **[2]**	**1.1a** **1.1**	Allow for *suvat* equation(s) used leading to a value for v or v^2 Allow sign errors Allow even if the sign of u does not match the sign of s and a
9	**(b)**		Using $v^2 = u^2 + 2as$ with $s = 0.03, u = \sqrt{51.68}, v = 0$ $0^2 = 51.68 + 2 \times 0.03a$ $a = -861.3...$ m s^{-2} N2L for pebble (downwards positive) $0.04g - R = 0.04a$ $0.04g - R = -0.04 \times 861.3$ $R = 34.8$ N	**M1** **A1** **M1** **A1** **A1** **[5]**	**3.1b** **1.1** **3.1b** **1.1** **1.1**	Allow for *suvat* equation(s) used leading to a value for a Allow for $s = 3$ used. FT their (a) Allow sign errors Need not be evaluated Use of N2L allow one error or omission Fully correct equation FT their acceleration. Weight must be included. Must be rounded to 3 sf. Accept 34.8 or 34.9 only
			Further Maths students may attempt an energy method for (b) Initial KE $= \frac{1}{2}mv^2 = \frac{1}{2} \times 0.04 \times 51.68$ GPE $= 0.04 \times 9.8 \times 0.03$ Work done against R is 1.04536 $R = \frac{1.04536}{0.03} = 34.8$ N	**M1** **A1** **A1** **M1** **A1**		Attempt to calculate change in KE or GPE Allow if GPE is not included